

Adapting Small Language Models to Italian Public Administration via KG-Augmented QLoRA Fine-Tuning

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Abstract

Public Administration (PA) regulations are characterized by high complexity, strict terminological precision, and interconnected legal frameworks. Standard Retrieval-Augmented Generation (RAG) systems often fail to capture the deep structural logic inherent in these texts. This project presents a complete end-to-end pipeline that adapts a Small Language Model (Qwen3-1.7B) to the Italian public procurement domain (Regione Puglia regulations). We integrate Parameter-Efficient Fine-Tuning (QLoRA) with a Hybrid Retrieval architecture that merges dense vector similarity with Knowledge Graph (KG) entity matching. Furthermore, we introduce an adaptive CrossEncoder reranking mechanism that uses an empirical threshold to detect unanswerable queries, gracefully falling back to the model’s parameterized knowledge to prevent hallucinations. Our ablation study demonstrates that QLoRA fine-tuning yields a **+24.8%** improvement in ROUGE-L and a **+41.7%** increase in Faithfulness over the zero-shot baseline. Notably, the hybrid KG retrieval slightly underperformed pure dense retrieval in Context Recall, revealing that automated entity extraction from complex legal prose remains a critical open challenge.

Keywords: Small Language Models, QLoRA, Knowledge Graphs, Retrieval-Augmented Generation, Legal NLP, Public Administration.

1 Introduction and Motivations

The integration of Large and Small Language Models into the Public Administration (PA) domain offers significant potential for simplifying access to complex regulatory frameworks. However, standard models suffer from hallucinations and a lack of specific, up-to-date regional knowledge [1]. While

Retrieval-Augmented Generation (RAG) [9] addresses this by fetching relevant text chunks, a pure vector-based approach struggles with tasks requiring multi-hop reasoning or strict terminological adherence [5].

In the context of the *Empulia* platform (the central purchasing body for Regione Puglia), standard vector-only systems often retrieve documents that are semantically similar but legally irrelevant. Furthermore, when the system fails to retrieve any useful document for out-of-domain queries, it is prone to hallucinate non-existent laws.

To address these challenges, we propose a comprehensive methodology combining:

1. **QLoRA Fine-Tuning** of a 1.7B parameter model to adapt to the formal tone of Italian legal texts.
2. **Hybrid Retrieval** leveraging both FAISS semantic search and a deterministic Knowledge Graph.
3. **Adaptive Fallback** utilizing a CrossEncoder reranker to threshold retrieved context relevance.

2 Related Work

2.1 Knowledge Graphs and Hybrid Retrieval

Recent literature highlights the critical limitations of purely vector-based retrieval in complex, specialized domains. Edge et al. [5] introduced GraphRAG, demonstrating that graph-based indices vastly outperform baseline RAG in multi-hop sense-making tasks over large corpora. Building upon this paradigm, Rahman et al. [11] established that hybrid retrieval systems—combining TF-IDF with dense neural embeddings—are essential for processing large technical documents where semantic drift often degrades accuracy. Their system achieved 68% reasoning accuracy with the Gemma 3 1B model, proving that combining statistical and neural retrieval with formula-preserving chunking maintains critical domain integrity. Our work adopts this hybrid paradigm by implementing a deterministic Knowledge Graph to anchor legal entities to their statutory roles alongside semantic vectors.

2.2 Small Language Models for Domain Reasoning

The specific application of Small Language Models (SLMs) for domain-level reasoning is proving highly efficient when combined with robust retrieval mechanisms. Bharadwaj and Jain [2] benchmarked RAG-assisted SLMs on the MMLU Pro dataset, demonstrating that while the 3B parameter LLaMa model benefited significantly from retrieved context (e.g., +18.5% accuracy in Organic Chemistry), poorly aligned context actually degraded

the reasoning of the larger 8B model by up to 20.25%. To overcome these inherent capacity limits and context-sensitivity issues, Pai and Hsu [10] investigated parameter-efficient fine-tuning (PEFT) for industrial RAG applications. Their findings confirm that domain-specific QLoRA fine-tuning of SLMs (such as DeepSeek-R1-Distill-Qwen-1.5B) dramatically reduces hallucination rates—increasing perfectly factual responses from 58.4% to 84.4%—while achieving high cost-efficiency (+0.53 F1 score improvement per GPU-hour).

2.3 Hallucination Mitigation and Iterative Refinement

While fine-tuning and hybrid retrieval mitigate many failure modes, residual hallucinations persist when processing out-of-domain queries. Iacob et al. [7] addressed this by proposing KGiRAG, an iterative feedback-driven GraphRAG architecture evaluated on the HotPotQA dataset. By utilizing an LLM-as-a-judge to incrementally expand the graph context until a strict factuality threshold (0.8) is met, they achieved a peak faithfulness of 0.95 and a completeness of 0.86 after three iterations, significantly outperforming static single-shot methods. Inspired by the need for such stringent quality-control mechanisms without the latency of iterative generation, our pipeline integrates an adaptive CrossEncoder reranker that empirically filters out irrelevant context entirely and defaults to parameterized knowledge, acting as an efficient safeguard against hallucination.

3 Proposed Approach

To address the challenge of domain-specific legal reasoning, the proposed system is architected as a sequential pipeline that progressively narrows the search space and refines language generation. First, raw regulatory documents are ingested and transformed into a dual-index structure—combining dense vector embeddings with a structural Knowledge Graph—to enable precise, multi-hop Hybrid Retrieval. Second, when a user submits a query, an adaptive CrossEncoder evaluates the retrieved context to filter out irrelevant information, acting as a routing mechanism that defaults to the model’s parameterized knowledge if the query falls outside the legal domain. Finally, the retrieved, highly-relevant context is injected into a prompt for our Small Language Model (Qwen3-1.7B). Because the SLM was fine-tuned via QLoRA specifically on Italian legal discourse, it can seamlessly synthesize the retrieved chunks into an accurate, formal, and legally sound response without hallucinating.

Retrieval-Augmented Generation (RAG) Architecture with Hybrid Retriever and Reranking

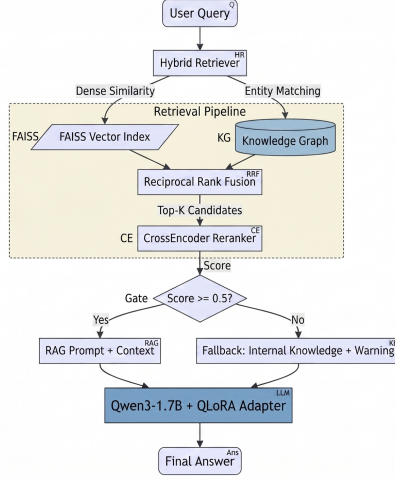


Figure 1: High-level architecture of the KG-Augmented QLoRA Hybrid GraphRAG pipeline.

3.1 Dataset and Preprocessing

The source corpus consists of 16 PDF documents representing the regulatory framework of *Empulia* (e.g., L.R. 37/2014, D.Lgs 50/2016). Text is extracted and processed using **Semantic Chunking**. Let a document D be a sequence of characters. We partition D into overlapping chunks $C = \{c_1, c_2, \dots, c_n\}$ using a maximum length $L = 1024$ and an overlap $O = 128$, ensuring logical paragraph boundaries are respected. This yielded 6,239 chunks.

Synthetic Dataset Generation: Due to the lack of pre-existing annotated datasets for the Regione Puglia legal framework, a synthetic QA dataset of 592 pairs was generated via a teacher-model approach to facilitate Instruction-Tuning (SFT) and quantitative evaluation. Gemini 2.0 Flash was prompted with few-shot examples to act as an Italian legal expert and generate diverse question-answer pairs based on a representative subset of the chunks. These pairs encompassed multiple cognitive levels, including factual extraction, Boolean verification, and complex synthesis across multiple legal articles. This dataset was then rigorously split into a training set (473 pairs) used for the QLoRA optimization, a validation set (59 pairs) to monitor convergence, and a strict, held-out test set (60 pairs) reserved exclusively for final evaluation.

3.2 Semantic Retrieval and Knowledge Graph

To combat the “semantic drift” of dense retrieval, we employ a Hybrid Retriever.

Dense Retrieval: Chunks are embedded using `paraphrase-multilingual-MiniLM-L12-v2` [8] into vectors $v \in \mathbb{R}^{384}$, stored in a FAISS index for efficient approximate nearest-neighbour search. The relevance of a chunk c to a query q is calculated via **Cosine Similarity**:

$$\text{sim}(q, c) = \frac{\mathbf{v}_q \cdot \mathbf{v}_c}{\|\mathbf{v}_q\| \|\mathbf{v}_c\|} \quad (1)$$

Knowledge Graph Retrieval: A lightweight Knowledge Graph is constructed via rule-based regex extraction over four entity categories: *articolo* (article references such as “Art. 36”), *legge/decreto* (statutory identifiers such as “D.Lgs. 50/2016”), *ente* (institutional bodies such as “ANAC” or “stazione appaltante”), and *concetto* (domain concepts such as “subappalto” or “aggiudicazione”). Entities co-occurring within the same chunk are linked as subject–predicate–object triples, enabling 1-hop graph traversal at query time. At inference, entities are extracted from the query q and matched against this KG to retrieve structurally related chunks.

Reciprocal Rank Fusion (RRF): The results from the dense and graph retrievers are fused. If $R_D(c)$ and $R_G(c)$ are the ranks of chunk c in the respective lists, the RRF score [3] is:

$$\text{RRF}(c) = \frac{1}{k + R_D(c)} + \frac{1}{k + R_G(c)} \quad (2)$$

where $k = 60$ is a smoothing constant.

3.3 CrossEncoder Reranking and Fallback

A CrossEncoder model (`ms-marco-MiniLM-L-6-v2`) re-evaluates the top- k retrieved chunks by processing the query and chunk jointly as a single input pair: $s = \text{CrossEncoder}(q, c)$. We implement an empirical threshold $\tau = 0.5$. If the maximum rerank score $\max(s) < \tau$, the system classifies the query as out-of-domain. Instead of injecting irrelevant context, it falls back to the model’s internal parameterized knowledge and alerts the user, mitigating hallucinations on out-of-scope queries.

3.4 QLoRA Fine-Tuning

To adapt the Qwen3-1.7B model to Italian legalese, we apply **QLoRA** [4]. The base model weights $W \in \mathbb{R}^{d \times k}$ are frozen and quantized to 4-bit NormalFloat (NF4). Trainable low-rank matrices $B \in \mathbb{R}^{d \times r}$ and $A \in \mathbb{R}^{r \times k}$ are

injected, modifying the forward pass:

$$h = Wx + \Delta Wx = Wx + \frac{\alpha}{r}BAx \quad (3)$$

This allows us to train only $\sim 1.0\%$ of the total parameters (17.4M out of 1.7B) while achieving full-parameter fine-tuning performance.

4 Evaluation and Results

4.1 Evaluation Metrics

We evaluated the system on the aforementioned held-out test set of 60 QA pairs. Because these 60 questions were strictly isolated and completely unseen during the QLoRA training phase, they serve as a robust, unbiased benchmark for measuring the true generalization capabilities of the architecture. We employed the **RAGAS framework** [6] (using GPT-3.5-turbo as the default LLM judge) alongside standard generative metrics. No repeated-run variance analysis was performed due to the computational cost of full-pipeline evaluation; the reported scores are therefore single-run point estimates.

- **ROUGE-L**: Measures the longest common subsequence between the generated answer and ground truth.
- **BERTScore**: Measures semantic similarity using contextual embeddings.
- **Context Recall**: Measures the proportion of ground-truth relevant sentences present in the retrieved context.
- **Context Precision**: Measures whether the retrieved chunks that appear early in the ranking are actually relevant to the ground-truth answer.
- **Answer Relevancy**: Measures whether the generated answer directly addresses the original question.
- **Faithfulness**: Measures whether the generated answer can be logically inferred purely from the retrieved context.

4.2 Training Convergence and Hyperparameters

A grid search over four configurations (LoRA rank $r \in \{8, 16\}$ crossed with learning rate $lr \in \{1 \times 10^{-4}, 2 \times 10^{-4}\}$) determined that learning rate is the dominant factor. The optimal configuration used $lr = 2 \times 10^{-4}$ with a Cosine decay schedule and a LoRA rank $r = 16$ ($\alpha = 32$), training 17.4M parameters. The final training ran for **3 Epochs**. Convergence analysis (Figure 2) showed a 40% training loss reduction, but after ≈ 40 steps, validation loss plateaued. This indicated that 3 epochs were the sweet spot to learn the domain without memorizing the 473 training samples.

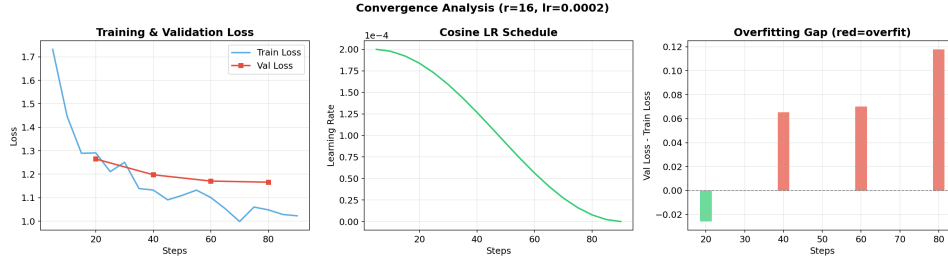


Figure 2: Training and Validation loss over 3 epochs, illustrating optimal convergence before the onset of overfitting.

Table 1: Ablation Study Results on Held-Out Test Set

Configuration	ROUGE	BERTScore	Recall	Faithful
Base + Dense	0.129	0.564	0.425	0.295
Base + Hybrid	0.111	0.551	0.418	0.256
QLoRA + Dense	0.161	0.654	0.425	0.418
QLoRA + Hybrid	0.149	0.658	0.418	0.389

4.3 Ablation Study

A 2x2 factorial ablation study (Table 1 and Figure 3) isolated the independent effects:

- **Fine-Tuning Impact:** QLoRA is the primary driver of performance. It drove a **+24.8%** increase in ROUGE-L, a **+15.9%** increase in BERTScore, and a massive **+41.7%** increase in Faithfulness over the baseline. The model successfully learned to synthesize answers in formal Italian rather than copying text verbatim.
- **Retrieval Impact:** Surprisingly, the Hybrid (KG) retrieval slightly underperformed the pure Dense retrieval (Context Recall dropped from 0.425 to 0.418). This negative finding suggests that the extracted Knowledge Graph may be sparse, causing Reciprocal Rank Fusion to occasionally down-rank highly relevant semantic chunks in favor of noisy graph matches.

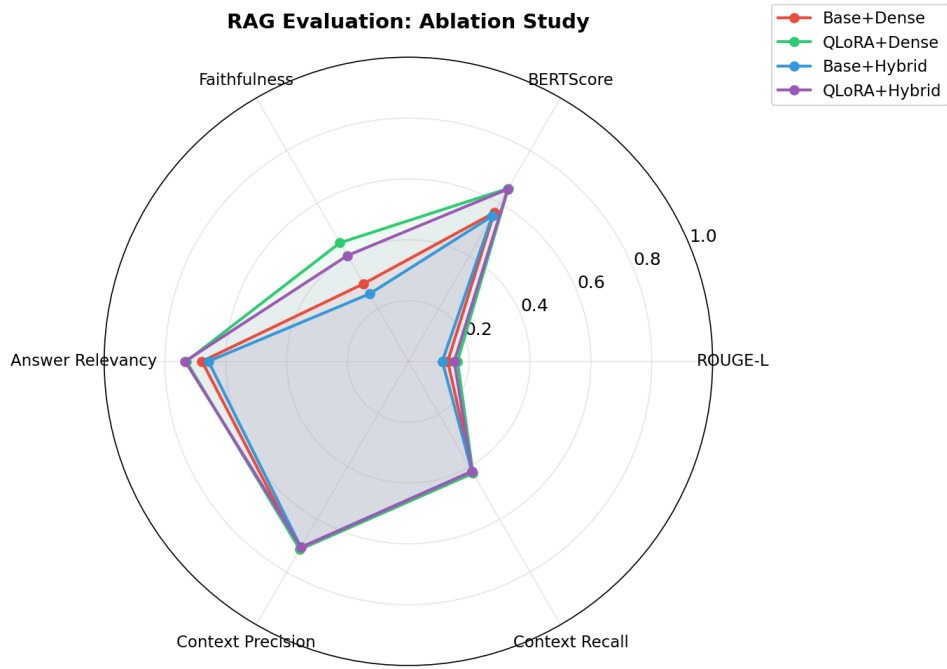


Figure 3: Radar chart comparing all four ablation configurations across six metrics. QLoRA fine-tuning dominates on generative metrics (ROUGE-L, BERTScore, Faithfulness), while retrieval mode differences are marginal. Context Precision and Answer Relevancy, also computed by RAGAS, are included here for completeness.

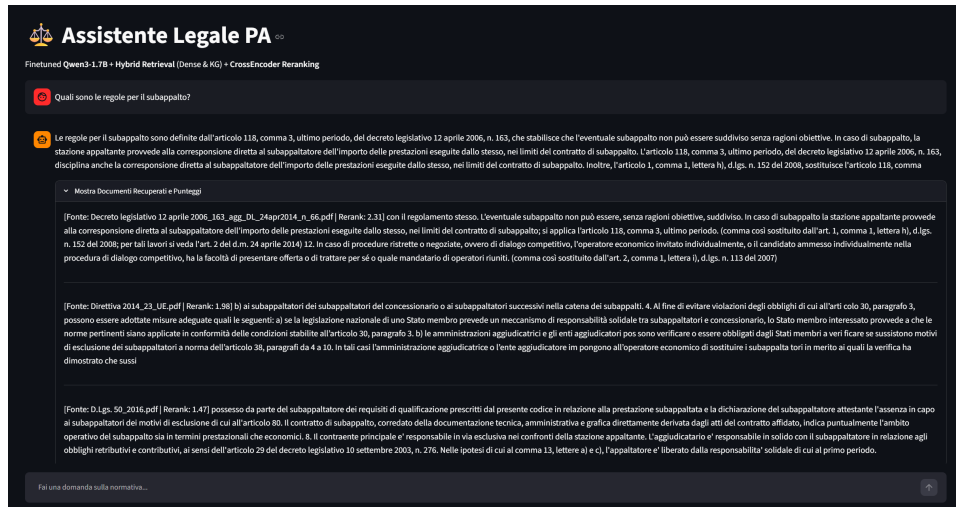


Figure 4: The local Streamlit application processing an in-domain query. The fine-tuned model leverages retrieved legal documents to provide an accurate, cited answer.

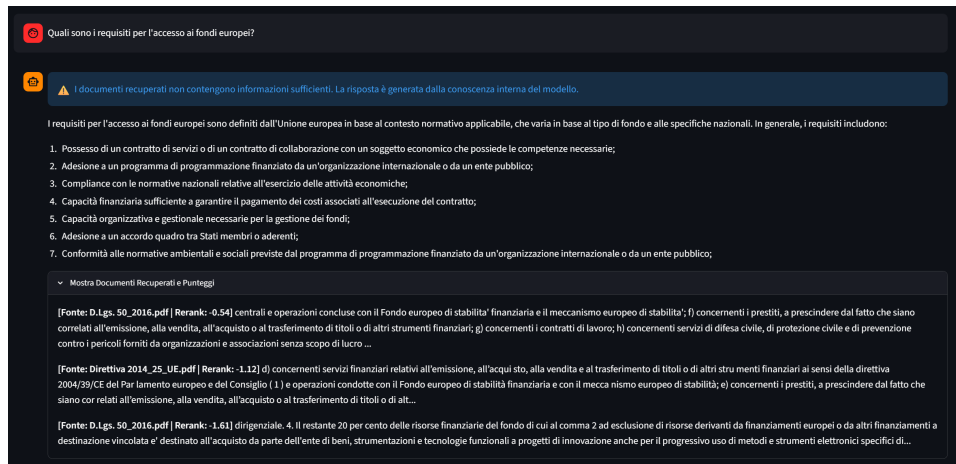


Figure 5: Adaptive Fallback mechanism: When the query is out-of-domain, the CrossEncoder score drops below 0.5. The system alerts the user and falls back to parameterized knowledge, avoiding hallucination on irrelevant documents.

4.4 Threshold Analysis

To evaluate the CrossEncoder fallback mechanism, we analyzed the score distributions for answerable versus unanswerable queries. As shown in Fig-

Figure 6, the ROC curve yields an AUC of 0.53, only marginally above the random-chance baseline of 0.50. This indicates that the CrossEncoder’s raw scores are a *weak* discriminator between in-domain and out-of-domain queries on our small test set, which is an acknowledged limitation. Nevertheless, the threshold $\tau = 0.5$ was selected empirically because, in practice, out-of-domain queries consistently produce rerank scores well below 0.5 while in-domain queries produce scores above 1.0, resulting in a functional separation despite the low AUC. This gap may widen with a larger and more diverse evaluation set.

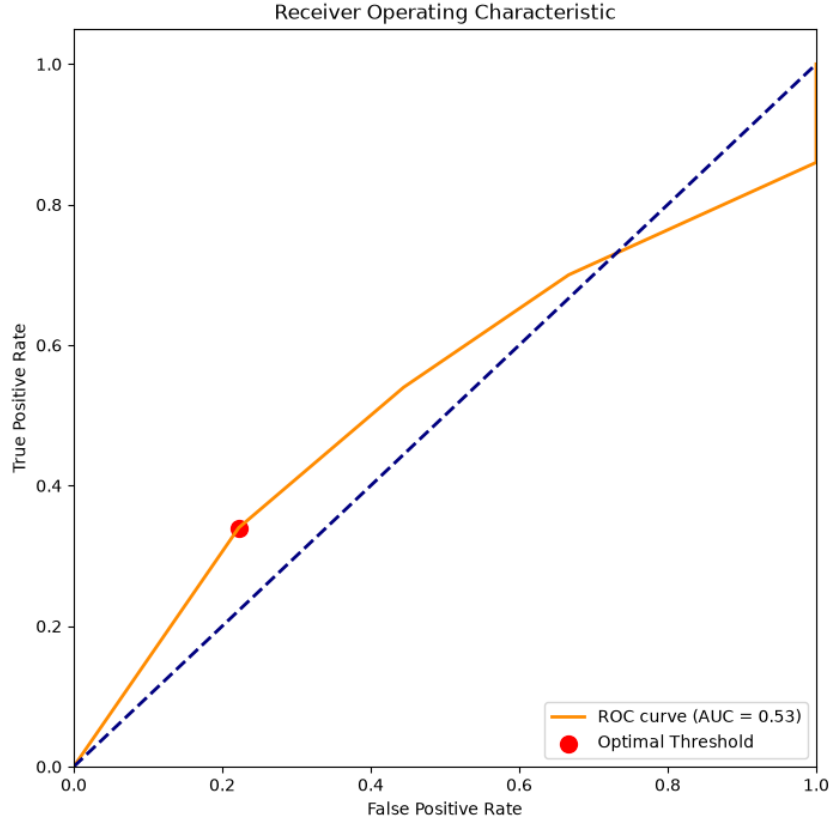


Figure 6: ROC Curve analysis determining the optimal CrossEncoder rerank score threshold ($\tau = 0.5$) for distinguishing answerable vs. unanswerable queries.

4.5 Real-World Application

The pipeline is deployed as a local Streamlit application (Figure 4). The CrossEncoder thresholding helps reduce hallucination risk on clearly out-of-domain queries. As shown in Figure 5, out-of-domain queries successfully trigger the *Knowledge Fallback* mode. No formal inference latency benchmarking was conducted; however, qualitative testing showed end-to-end response times of approximately 10–20 seconds per query on a consumer GPU via DirectML, which is acceptable for an interactive advisory tool but would require optimization (e.g., vLLM, TensorRT) for production-scale deployment.

5 Reproducibility and Setup

To ensure reproducibility, the full extraction, synthetic data generation, and QLoRA fine-tuning pipelines were executed on a single Kaggle **Nvidia Tesla T4 (15.6 GB VRAM)**. The local inference app runs on consumer-grade hardware accelerated via Windows DirectML, demonstrating the democratization of domain-specific SLMs.

6 Conclusions and Future Work

This project implemented an end-to-end Hybrid GraphRAG pipeline for the Italian Public Administration domain. The ablation study validated the effectiveness of the proposed methodology: QLoRA fine-tuning proved essential for adopting the required legal lexicon, while the CrossEncoder-based routing threshold provides a practical safety net against hallucination on out-of-domain queries.

However, it is important to contextualize the **absolute metric scores**. While the best configuration achieves a ROUGE-L of 0.161 and a BERTScore of 0.658, these figures cannot be directly compared to the high absolute accuracies reported in related literature (e.g., the 68% multiple-choice accuracy on MMLU Pro by Rahman et al., or the 0.95 faithfulness on Wikipedia-based HotPotQA by Iacob et al.). Our evaluation is strictly generative and operates within the Italian Public Administration domain—one of the most complex NLP environments due to its rigid terminology, nested clauses, and multi-hop legal dependencies.

In this highly constrained setting, the true measure of success is the *relative improvement*. QLoRA fine-tuning drove a massive **+24.8%** relative increase in ROUGE-L and a **+41.7%** increase in Faithfulness over the zero-shot baseline. These dramatic relative gains align with the findings of Pai and Hsu [10], proving that SLMs can successfully internalize complex domain logic and suppress hallucinations when properly adapted. Several factors cap

the absolute ceiling of our current metrics:

- **Model capacity:** Qwen3-1.7B, while efficient, has inherently limited reasoning and generation capabilities compared to models with 7B+ parameters. The compressed 4-bit quantization further constrains representational fidelity.
- **Knowledge Graph Sparsity:** Contrary to initial hypotheses, the Hybrid retrieval slightly underperformed pure dense retrieval in Context Recall. This highlights a critical limitation: automated KG extraction from complex legal text can be noisy and sparse. When fused via RRF, noisy graph matches occasionally down-ranked highly relevant semantic chunks.
- **Domain complexity:** Italian legal prose is characterized by exceptionally long sentences, nested subordinate clauses, and domain-specific terminology that poses a significant challenge for small generative models.

Despite these limitations, the architecture demonstrates a clear and consistent directional improvement across generative metrics, confirming that parameter-efficient fine-tuning is a viable strategy for resource-constrained, domain-specific NLP. Future work should explore: (i) scaling to larger base models (e.g., 7B–8B parameters) while preserving the QLoRA efficiency; (ii) improving the automated Entity Resolution pipeline to densify the Knowledge Graph; and (iii) implementing incremental Knowledge Graph construction to support real-time regulatory updates.

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